DO NOW — RECALL

Operating-system functions

Name: _____ Class: _____ Date: _____

1. What does LMC LDA do?

2. What does the CPU control unit do?

3. Define a process.

4. Binary search reduces a sorted 32-element candidate set to one. How many successive halvings is this?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Loads ACC from a mailbox.

2. Model answer

Coordinates and decodes execution.

3. Model answer

An instance of a program being executed or scheduled.

4. Model answer

5: 32→16→8→4→2→1.

Vocabulary — know these terms

process — Running/scheduled program instance

filesystem — Organised file storage

driver — Device interface software

spooling — Queued peripheral work

NEW LEARNING

Read, annotate and explain

Understand

The OS mediates application requests and access to hardware.

Trace

Resource management schedules processes, allocates memory, controls files and drivers.

Explain

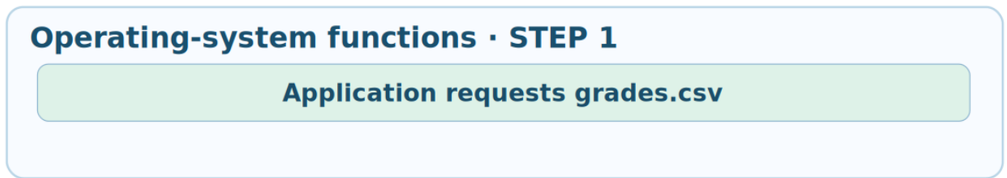
Permissions and isolation help keep processes and users from accessing unauthorised data.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

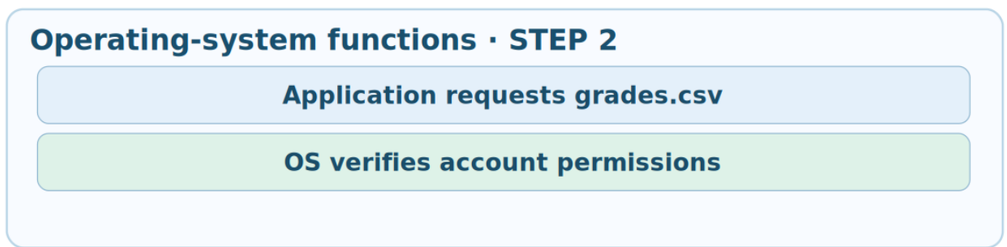
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



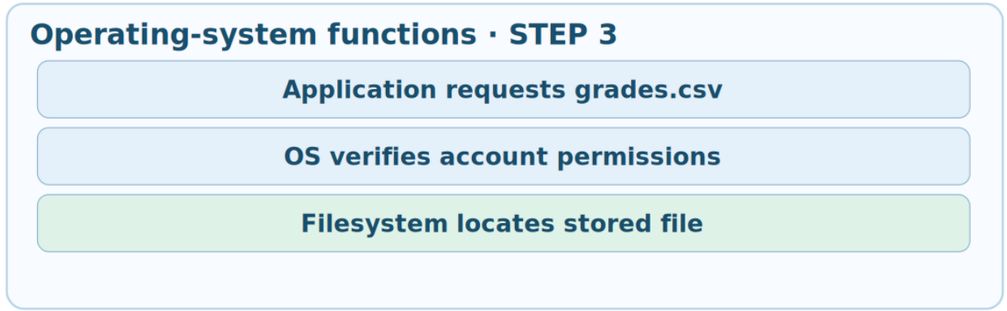
Operating-system functions · STEP 1

New Learning



Operating-system functions · STEP 2

New Learning



Operating-system functions · STEP 3

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

An app prints a document while a printer is busy. Explain the OS's role.

The print job enters a queue and is later sent via a compatible printer driver when the device is available.



WEDO · STATE TRACE

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

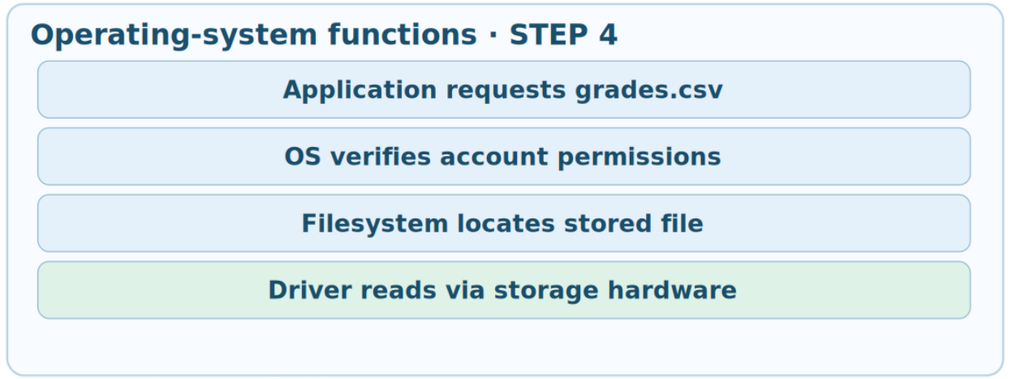
Guided 1 — CHECK

Predict first
The print job enters a queue and is later sent via a compatible printer driver when the device is available.

New Learning · original lesson diagrams

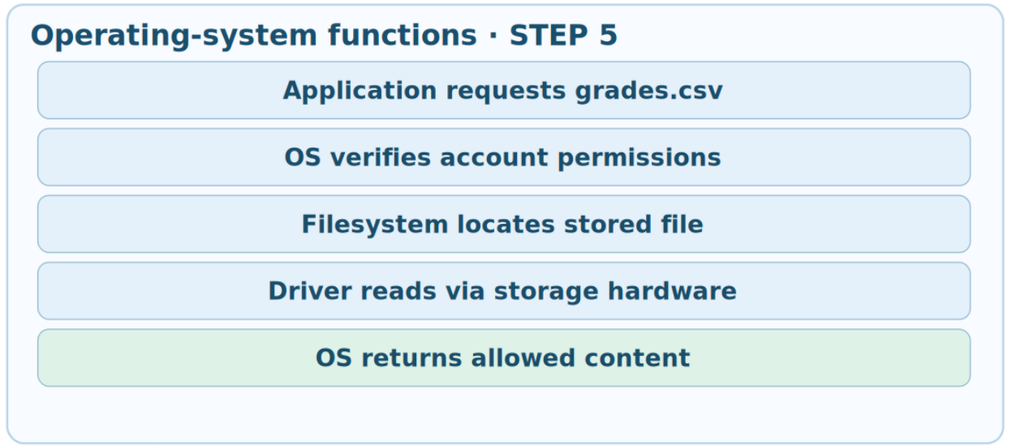
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Operating-system functions · STEP 4

New Learning



Operating-system functions · STEP 5

I DO — WORKED EXAMPLE

Follow the worked method

A pupil app requests a protected file. Explain the OS steps.

Predict first

The OS checks authorisation before returning content; a missing permission should prevent the read.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Applications must directly control all hardware to read files.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

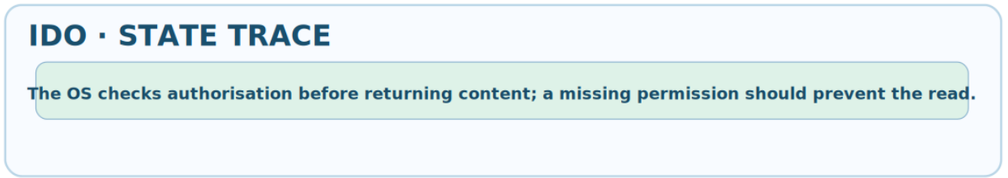
Explain how an OS manages a browser, game and backup concurrently, including protection. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A pupil app requests a protected file. Explain the OS steps.

The OS checks authorisation before returning content; a missing permission should prevent the read.



IDO · STATE TRACE

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

An app prints a document while a printer is busy. Explain the OS's role.
Source code / input:
Application → print spool → driver → printer

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

State four operating-system responsibilities. [4]

Question 2

Describe a protected file read from application request to result. [6]